When the Benchmark Grades Itself: Two Information Leaks in Football Player-Representation Evaluation

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Abstract

Learned representations of football (soccer) players are routinely reported to capture playing style, yet the field validates them in ways that leak the relevance signal into the model being graded. We identify two such leaks, give leakage-free protocols, and release an open benchmark and evaluator. First, a set-transformer over anonymized StatsBomb 360 freeze-frames scores 0.878 within-tournament identity Top-1 under a random split but collapses to 0.188 under a match-disjoint split (disjoint confidence intervals), with a same-match control at 0.889 isolating match-context rather than enduring style; the collapse reproduces for static heatmap representations (a parameter-free heatmap and a Player Vectors-style NMF), so it is a property of the split protocol, not one architecture, though we do not claim a discriminatively-trained network such as 6MapNet collapses identically. Second, models graded against their own clusters reach nDCG@10 of about 0.99, but on an external benchmark anchored to realized like-for-like replacement transfers, no learned representation we test beats a raw-statistics nearest-neighbour on all-candidates retrieval, the primary metric – a null the benchmark is powered to reject a 0.007-MAP advantage against. This holds for a faithful Player Vectors reproduction, a purpose-trained metric learner, and genuinely orthogonal richer inputs (event-transformer, VAEP, and 360 spatial features), and replicates across two disjoint player pools; within-role retrieval is near-floor and underpowered, so we confine claims to all-candidates. We also correct a dependence error in the field's significance testing: about 1363 queries collapse to roughly 22 position components (effective count about 8), so per-query tests understate variance and the naive p is about ten times too small.

1 Introduction

Player-similarity search ("find a midfielder like X"), style fingerprinting, and representation learning over event and spatial data are all active areas of football analytics. A less examined question is how these representations are *evaluated*. We find two pervasive pitfalls, and in both cases the problem is that information about the relevance signal leaks into the very model being graded.

The first pitfall is circular relevance. Similarity quality is judged against labels derived from the model itself, such as k-means archetypes or role clusters, or else against anecdotes. A model graded on its own partition can score nearly perfectly; our own legacy text benchmark, whose relevance labels are the model's k-means clusters, yields nDCG@10 of about 0.99, while the same model fails any external test. The second pitfall is match-context leakage. Identity and consistency are scored by splitting a player's events at random within a tournament, so events from the same match land in both the query and the gallery. The model can then recognize "same game" rather than enduring style.

We instantiate an external, leakage-free protocol for each pitfall and report what learned representations actually achieve once the leak is removed. The two leaks share a single underlying cause, which we develop in Section 3: the evaluation's assumed independence between the relevance signal and the model under test is violated. They break that independence in different places, so each needs its own fix.

We treat the two contributions asymmetrically, and say so plainly. The label-leakage study (Pitfall 1) is the paper's primary contribution: it ships a full external benchmark, many baselines including a faithful published method, and a cross-pool generalization result. The feature-leakage study (Pitfall 2) is a focused cautionary result on one task and one dataset family, with a correspondingly narrower claim. The two-by-two view in Section 3 organizes the framework; it is not a claim that the two results carry equal evidential weight.

Our contributions are:

1. An evaluation-leakage framework that distinguishes *label leakage* (the relevance labels are produced by the model under test) from *feature leakage* (the query and gallery share nuisance input features), and shows that both are special cases of a violated independence assumption.

2. ScoutBench, an external, transfer-anchored similarity-retrieval benchmark, together with the resulting negative result: no learned representation we test, including a faithful reproduction of Player Vectors, beats a raw-statistics nearest-neighbour baseline, and the finding replicates across two disjoint player pools.

3. A cross-match player-identity protocol and the collapse it exposes: a set-transformer drops from 0.878 to 0.188 Top-1 once the query and gallery are drawn from disjoint matches, with a same-match control isolating match-context.

4. A dependence and significance correction. The field's default per-query bootstrap treats clustered queries as independent and inflates significance by roughly an order of magnitude; we develop the corrected block-bootstrap and cluster-level tests as a first-class methodological contribution.

2 Related Work

The methodological survey by Davis et al. [3] catalogs the evaluation protocols used across sports analytics, including the consecutive-season nearest-neighbour test for player similarity, and the broader pitfalls that attend them. It supports the framing we adopt here: in practice, player-similarity and representation work is validated either circularly, against labels the model itself produces, or qualitatively, through scouting case studies.

The pattern is visible across the representative learned representations. Player Vectors [4] builds per-type non-negative matrix factorizations over action-location heatmaps and is evaluated through consecutive-season player-identity retrieval, where the labels are objective, supplemented by qualitative scouting case studies; it does not appeal to external transfer ground truth. RisingBALLER [1], a players-as-tokens transformer foundation model, is evaluated quantitatively on next-match-statistics prediction, while its similar-player claims remain qualitative. SoccerMix [5], a soft-clustering mixture model for action and style, is evaluated qualitatively, as is Football2Vec [11], a 32-dimensional Word2Vec-style player embedding assessed through qualitative cosine nearest

Table 1: The two leakage mechanisms by task family, with the fix that restores the broken independence assumption. The two results are not of equal weight: the label-leakage row is the paper's primary contribution, the feature-leakage cell a focused cautionary result.

Leakage source	Similarity retrieval	Identity / consistency
Label leakage	self-derived archetype relevance, nDCG about 0.99; fix: external transfer anchor (ScoutBench)	n/a (identity labels are objective)
Feature leakage	mitigated (candidates are other players)	within-tournament split-half, 0.88; fix: match-disjoint split

neighbours. A recent spatial-statistics approach, the spatial similarity index of Gómez-Rubio et al. [7] (Lee's statistic and Moran's I over kernel-smoothed position heatmaps), is likewise demonstrated only through illustrative case studies, with no external relevance ground truth and no released benchmark. Across this body of work, similarity is therefore validated either against the model's own structure or by inspection, never against an external outcome.

Our work sits in the genre of method-level reality checks. A Metric Learning Reality Check [12] showed that reported gains in deep metric learning largely evaporate under fair protocols, and the tabular-learning literature has repeatedly found that gradient-boosted trees match or beat deep models on typical tabular data [13, 8]. Dimensional collapse in contrastive learning [9] offers a plausible mechanism for why a learned low-dimensional projection can be lossy relative to its raw inputs. Our raw-versus-learned result is the football-similarity instance of these findings.

External-outcome evaluation does exist in football analytics, but it targets quantities other than similarity. Market-value prediction [2] forecasts a scalar value rather than a similarity ranking, and transfer-impact forecasting [6] predicts the effect of a move. Neither uses realized like-for-like replacements as a similarity-retrieval ground truth, which is the precise gap ScoutBench fills. The closest neighbour to our identity task is 6MapNet [10], a triplet network for player identity from tracking data. It splits its data by sampling player phases per entity into train, validation, and test, rather than by match, so phases from a single match can fall into both query and gallery; it may therefore be vulnerable to the same match-context leakage we study, and we treat it as a candidate to re-evaluate under a match-disjoint split rather than as a safe reference.

3 A Unifying View: Evaluation Leakage

Both pitfalls are, at bottom, one failure: the evaluation's assumed independence between the relevance signal and the model under test is violated. In label leakage, the relevance labels are produced by the model itself, through its own clusters or archetypes, so the model is graded against its own output and scores near-perfectly but meaninglessly. In feature leakage, the query and the gallery share input features drawn from the same match, so the model can match shared nuisance context instead of the target property of enduring identity or style. Table 1 arranges the two mechanisms against the two task families.

These are two distinct contamination mechanisms, not one mechanism described twice. Label leakage violates label-model independence, because the relevance labels are generated by the model under test. Feature leakage violates query-gallery independence, because the query and gallery share nuisance match-context beyond the target property, even though the identity label itself is objective and independent of the model. Valid retrieval evaluation requires both conditions to hold;

each pitfall breaks a different one, and each fix restores the broken one, an external label anchor for the first and a match-disjoint split for the second. The shared lesson is not a single mechanism but a discipline: identify every path from the relevance signal to the representation, and sever it.

4 Pitfall 1: Circular Relevance, and ScoutBench

4.1 The benchmark

ScoutBench Task B grounds similarity in realized like-for-like replacement transfers drawn from CC0 transfermarkt data. When a club loses player X at sub-position P and signs player Y at P in the same or next window, the pair (X, Y) is taken as a *silver* similarity-positive, that is, a label that is objective and outcome-derived but noisy rather than a clean gold-standard judgment. Each query player is ranked against all others by embedding cosine similarity. We report MAP, Hit@k, and MRR over all candidates as the primary metric, and over same-sub-position candidates as a stratified secondary metric. There are 1363 query players drawn from 5868 pairs, against a gallery of 5668 players derived from StatsBomb open data and used for research only.

4.2 Significance done right

The query-to-relevant graph collapses into 22 connected components, roughly one per sub-position, so the queries are not independent. We therefore use a block bootstrap that resamples whole components ($B = 10000$) and, separately, a conservative cluster-level paired t-test over the 22 component means, with a Holm correction across the comparison family. The field's default per-query bootstrap, which we argue is wrong here, treats the 1363 dependent queries as independent. It understates variance by roughly a factor of 1.3 and makes the naive p-value about ten times too small, that is, it inflates significance by roughly an order of magnitude. We treat this correction as a methodological contribution in its own right. We also note that the query-weighted block bootstrap and the component-equal-weighted cluster t-test estimate different targets, the effective component count being about 8, so we report both rather than presenting one as a conservative version of the other.

4.3 Results

Table 2 reports the leaderboard. All-candidates MAP (ALL-MAP) is the primary metric; same-position MAP (SP-MAP) is the stratified within-role test, which we show below (Sec. 4.4) is near-floor and underpowered. The headline result, on all-candidates, is that no learned representation beats the raw-statistics nearest-neighbour baseline, while every method clears random.

A genuine signal does exist on all-candidates retrieval. The raw card beats random by 0.0163 ALL-MAP, which is significant under both the block bootstrap ($p = 0.0002$) and the conservative component t-test ($p = 0.0001$). On same-position retrieval the signal is weaker: it is significant under the block bootstrap ($p = 0.0002$) but not under the component t-test ($p = 0.15$), which is consistent with the position prior dominating within a role, since a random vector already reaches SP-Hit@10 of about 0.18. The robust signal-exists claim is therefore the all-candidates one, and the same-position regime sits near the floor.

The null is well-powered on the primary metric and near-floor on the stratified one, which we establish by measuring the minimum detectable effect (MDE) of the conservative component t-test directly from the per-component variance (power 0.8, $\alpha = 0.05$; `scoutbench_power.py`). On all-candidates the MDE is about 0.007 MAP, below the best-minus-random spread of 0.0163, so the

Table 2: ScoutBench Task B leaderboard. SP-MAP is same-sub-position MAP; ALL-MAP is all-candidates MAP (the primary metric). Higher is better.

Method	SP-MAP	ALL-MAP
fbref percentile	0.0514	0.0082
pca(card)	0.0501	0.0195
raw_card cosine	0.0498	0.0194
TF-IDF text	0.0469	0.0165
Player-Vectors (faithful, Decroos & Davis heatmap-NMF)	0.0464	0.0172
NMF (over card features)	0.0450	0.0135
v11 (contrastive embed)	0.0447	0.0162
card+VAEP	0.0440	0.0088
random	0.0369	0.0030

test resolves spread-sized effects: it detects raw beating random (component difference +0.0097, $p \approx 0$) and would register a learned-representation advantage of 0.007 MAP or more, yet the learned methods sit at or below raw (raw minus v11 +0.0008, $p = 0.49$; raw minus Player Vectors −0.0027, $p = 0.47$). The all-candidates null is therefore genuine, not a failure of power. On same-position the MDE is about 0.017 MAP, exceeding the entire 0.0145 spread, so the within-role test cannot resolve even raw beating random ($p = 0.15$) because the position prior dominates. We accordingly draw no conclusions from same-position differences, including the apparent raw-minus-v11 gap; they fall below the label's resolution. This is itself the lesson: circular self-labels manufacture an effect (nDCG of about 0.99) far above this floor, which is precisely why they look near-perfect.

The off-objective contrastive embedding does not beat raw. Here "off-objective" means the embedding was trained on a different objective than replacement retrieval. On all-candidates, the primary and well-powered metric, v11 sits at or below raw (raw minus v11 +0.0008, component t-test $p = 0.49$), a powered null. The nominal raw-minus-v11 same-position gap of 0.0051 is significant only under the query-weighted block bootstrap (p about 0.02); it does not survive the Holm correction and is not robust to the component t-test ($p = 0.36$), and given the same-position MDE of about 0.017 it falls below the benchmark's resolution, so we rest no claim on it.

The faithful, published Player-Vectors baseline also fails to beat raw, which is the key external-baseline check. Our reproduction of Player Vectors [4] builds 50-by-50 action-location heatmaps per type (pass start and end, and shot, cross, and dribble starts), applies per-type NMF, draws purely on on-ball events, covers the full gallery of 5668 players, and uses none of our card features. It scores SP-MAP 0.0464 and ALL-MAP 0.0172. It is not significantly different from the raw card: the raw-minus-PV gap is 0.0034 SP-MAP (block $p = 0.15$, component t $p = 0.90$) and 0.0022 ALL-MAP ($p = 0.15$). The canonical published learned representation thus ties or slightly trails a raw-statistics nearest-neighbour, which moves the claim from "our models lose" to "learned representations, including the published Player-Vectors method, do not beat raw." A secondary NMF over the card features behaves the same way, scoring SP-MAP 0.0450, below raw on the query-weighted estimand ($p = 0.003$) but not robust to the component-level multiplicity correction.

Purpose-trained, player-disjoint metric learning does not significantly beat raw, which is the decisive, non-tautological test. We train a projection directly on the replacement objective using a multi-positive InfoNCE loss that masks each player's other true replacements, since the replacement graph is many-to-many and a naive one-positive-per-row loss would create contradictory labels. Evaluated on player-disjoint held-out queries across three seeds, the same-position MAP gap of the metric learner minus raw is +0.0008, +0.0006, and −0.0027, all with $p > 0.37$, while the all-candidates gap is $+0.0016$, $+0.0024$, and $+0.0009$, consistently small and positive but never significant (p between 0.19 and 0.68). A predeclared TOST equivalence test, that is, two one-sided tests at a ± 0.005 MAP margin, concludes statistical equivalence to raw in one of the three seeds and is inconclusive in the other two, where the 95% confidence intervals marginally exceed the margin, for example $[-0.006, +0.007]$. The honest statement is that there is no improvement over raw in any seed, and equivalence is only partially established: a metric learned directly on these features does not beat raw cosine, though we cannot fully rule out a sub-0.005-MAP gain.

Richer, orthogonal inputs also do not beat raw on the primary metric, which is the strongest test of the claim. Every representation above only reshapes the 88-dimensional card, so we additionally test inputs the card does not contain: a 256-dimensional event-transformer embedding over SPADL event sequences (full coverage, with similarity-structure Pearson correlation of about 0.01 against the card, hence genuinely orthogonal), a 3-dimensional VAEP representation, and a StatsBomb-360 spatial fingerprint on a 923-player tournament subset. Fused with the card via training-free cosine nearest-neighbour, none beats the raw card on all-candidates (card+event SP-MAP 0.0503 against raw's 0.0498; card+VAEP and card+event+VAEP at or below raw; card+360, on its subset, 0.0936 against raw's 0.1007), and a metric MLP trained over the fused inputs (player-disjoint, three seeds) likewise does not (SP-MAP gap $+0.004 \pm 0.004$, all seeds non-significant). The one positive lead, reported on the same footing as every other claim, is the event embedding alone within-position: SP-MAP **0.0703 against raw's 0.0498** (block $p = 0.0002$), the largest within-role effect we observe. We do not count it as a win, but for exactly the reason the within-role regime decides nothing: it fails the component t-test ($p = 0.11$) and does not transfer to all-candidates (0.0093 against 0.0194), held to the identical near-floor, underpowered standard (same-position MDE about 0.017) we apply to raw-versus-random and raw-versus-v11, not a stricter one. Event sequences are thus the most promising direction for within-role discrimination and the obvious next experiment, not noise to be dismissed; they simply do not clear the bar on the metric the benchmark can adjudicate.

Finally, the circular alternative makes the contrast concrete. The same v11 model scores nDCG@10 of about 0.99 on a benchmark whose relevance is its own k-means labels. External grading reverses the verdict.

4.4 Honest scope of the negative result

The absolute numbers are low. Every method retrieves the realized replacement rarely, with same-position Hit@10 of about 0.27, and a random vector already reaches 0.175 there because the position prior dominates. The contribution is the ordering and the external protocol, not deployable accuracy. The safe claim is that no learned representation we test beats a raw-statistics nearest-neighbour on all-candidates external transfer-anchored similarity — a null the benchmark is powered to overturn (MDE about 0.007, below the method spread), not a large-margin positive result; we confine it to all-candidates because same-position is near-floor and underpowered. The faithful Player-Vectors reproduction makes this a claim about a published learned method, not only about our own models. Two baselines remain for future work: RisingBALLER, a transformer foundation model requiring days of training, and 6MapNet's exact triplet convolutional network under a cross-match split, which needs tracking data we lack — we test the static heatmap family it builds on under P2 below.

4.5 Generalization across player pools

We re-run Task B independently on two disjoint pools, split by the league from which the card was built. Pool A is the top-five men's club leagues, with 3049 players and 756 queries; Pool B is

Table 3: Within-tournament player-identity Top-1 and MRR by split condition, with 95% player-level bootstrap confidence intervals. The split-half and cross-match intervals are disjoint.

Condition	Top-1 [95% CI]	MRR
split-half (random, standard)	0.878 [0.846, 0.905]	0.924
same-match control (disjoint halves of one match)	0.889 [0.859, 0.917]	0.932
cross-match (disjoint matches, leakage-free)	0.188 [0.153, 0.224]	0.326
cross-match, no teammate tokens	0.137 [0.107, 0.171]	0.274
reference: cross-tournament / random	0.178 / 0.007	—

international tournaments together with women's competitions and the ISL and others, with 2619 players and 479 queries. The conclusion replicates in both. In Pool A no learned representation beats raw: raw scores 0.0639 against v11's 0.0637 (non-significant) and Player-Vectors' 0.0598 (non-significant). In Pool B the same holds: raw scores 0.0841 against v11's 0.0648 and Player-Vectors' 0.0677, with neither learned method ahead. Raw beats random in both pools, by 0.0182 in Pool A (block $p < 0.001$ and component t $p = 0.003$, the cleanest signal-exists result) and by 0.0222 in Pool B (block $p = 0.007$). The finding is therefore not an artifact of a specific pool. Absolute MAPs are higher within each homogeneous pool, since replacement retrieval is easier among more similar candidates.

5 Pitfall 2: Within-Tournament Match-Context Leakage in 360 Identity

5.1 Task and protocol

The task is player-identity retrieval from anonymized 360 freeze-frames. For each on-ball event we form up to 23 player point-tokens, each carrying an (x, y) location and an actor, teammate, or opponent role. A set-transformer is trained contrastively and evaluated on held-out players, with 701 players for training and 350 for test, over 528,059 events drawn from World Cup 2022, Euro 2024, and Euro 2020 across 166 matches. The model is identical across all conditions; only the query and target split changes.

5.2 Results

Table 3 reports the conditions. The confidence intervals are player-level bootstraps that resample held-out players. The split-half and cross-match intervals are disjoint, so the collapse is unambiguous. It is not a single-pool artifact: the macro-average that weights each of the three tournaments equally matches the pooled number (split-half 0.88, cross-match 0.19), and every tournament collapses, with cross-match per-tournament Top-1 of 0.24, 0.15, and 0.19 against split-half values of 0.95, 0.86, and 0.83.

The collapse itself is large. Top-1 falls from 0.878 under split-half to 0.188 under cross-match, a drop of 0.69, at comparable event volume, about 137 against 141 events per side, so it is not a data artifact. Within-match information is near-sufficient: the same-match positive control scores 0.889, so when the query and target share a match, identity is nearly perfect, and when they do not, it collapses. One caveat applies, however. The control halves a single match, leaving about 71 events per side, so it is not volume-matched to the split-half and cross-match conditions at about

140 events per side; it isolates the presence of match-shared information rather than a volume-controlled effect. The defensible conclusion is that random within-profile splitting substantially exploits match-specific information, and we cannot cleanly separate match identity from team, lineup, and role-within-that-match from genuine short-term individual behaviour.

A small leakage-free signal nonetheless survives. Cross-match Top-1 of 0.188 greatly exceeds the random baseline of 0.007, dropping to 0.137 once teammate tokens are removed. Some individual or structural signal persists, but its interpretation as individual style is bounded by the confounds named below.

The collapse is not specific to our architecture, with a bounded generalization claim. *Static* heatmap representations collapse the same way under a match-disjoint split. A parameter-free multi-channel heatmap (actor location, teammate density, and opponent density) drops from 0.806 (95% CI [0.768, 0.838]) under random split-half to 0.113 ([0.083, 0.146]) cross-match; a Player Vectors-style NMF reduction drops from 0.595 ([0.549, 0.639]) to 0.123 ([0.093, 0.155]). Both show disjoint split-half and cross-match intervals and a high same-match control. The leak is therefore a property of the within-tournament random-split protocol, not of any single architecture: it strikes a parameter-free histogram and an NMF reduction exactly as it strikes our trained set-transformer. An important scope caveat is that these are static, untrained representations. We do *not* claim a discriminatively-trained triplet convolutional network such as 6MapNet collapses identically, since a learned model could in principle acquire match-invariant features a static histogram cannot. What we show is that the protocol leaks for the heatmap input family on which published methods rest — Player Vectors directly, and 6MapNet's heatmap inputs — while 6MapNet's exact network needs continuous tracking data we lack and remains untested.

The implication should be stated carefully, and we do not accuse the literature. Within-tournament random split-half is a tempting, seemingly-natural protocol that would inflate identity by roughly 4.7 to 6 times through match-context leakage. The established literature, however, does not obviously use random within-tournament split-half. Player Vectors [4] evaluates consecutive-season consistency, which is match-disjoint with respect to exact same-match overlap, though team and role context can still recur across seasons. 6MapNet [10], the closest neighbour, samples player phases, potentially several from one match, into train, validation, and test rather than splitting by match; and the static heatmap input family it builds on collapses under our match-disjoint split (above), so a phase-sampling protocol like 6MapNet's is exposed in principle. Whether its trained network actually collapses is untested — it needs tracking data we lack — so we flag it as a candidate to re-evaluate, not a proven case and not a safe reference. Our result is therefore a cautionary protocol finding together with a cheaper safe alternative: cross-match-within-season splitting recovers match-disjointness without requiring multiple seasons, and a same-match positive control should accompany any identity claim.

Two residual threats deserve naming, though neither is fatal. First, train-time leakage: the encoder was trained with random-half, same-match positives, so it may have learned to exploit match context, and a model trained with cross-match positives might recover signal. We therefore claim only that, as trained and evaluated, the 0.88 figure reflects match context, not that individual style is unlearnable from 360 data. Second, the no-teammate ablation at 0.137 also removes about half the input tokens, confounding model capacity with leakage type; a cleaner test would permute teammate identities rather than delete tokens, which we leave to future work.

6 Leakage-Free Protocols

We package the two fixes as reusable protocols. ScoutBench Task B is an external, transfer-anchored similarity-retrieval evaluation with an open evaluator, a block bootstrap paired with a cluster t-test and Holm correction, and all-candidates retrieval as the primary metric. The cross-match identity protocol is a match-disjoint query and target split, volume-comparable to split-half at about 140 events per side, accompanied by a same-match positive control. Both ship with code and corrected baselines, and both are cheap to adopt.

7 Recommendations

Three recommendations follow directly. First, never grade similarity against a model's own labels; use external or outcome anchors, and always report the random-baseline floor. Second, for identity and consistency, split by match or by season, never randomly within a competition, and include a same-match control. Third, treat queries as clustered into position components, use cluster-level inference, and report absolute floors, effect sizes, and the test's minimum detectable effect rather than p-values alone — a null is only informative if the benchmark could have detected a real advantage of the relevant size (here, well-powered on all-candidates, near-floor within-role).

8 Threats to Validity and Limitations

The replacement labels are silver: realized replacements encode budget, availability, and positional need, not pure similarity. We checked their validity directly. Realized replacements sit at a mean similarity-percentile of 0.549 among same-position players, with a sub-position-clustered confidence interval of $[0.521, 0.585]$ that excludes the 0.50 null, and replacement cosine of 0.449 against a random-same-position 0.404. The label therefore carries genuine but weak stylistic similarity signal beyond position; it is noisy rather than pure noise. This coheres the benchmark: a weak label signal explains why all methods beat random yet cluster near the floor and none separates cleanly, since the raw card captures the linearly-available similarity and the residual is mostly non-stylistic transfer noise.

The evaluation draws on a single dataset family, StatsBomb open and Wyscout public, which is non-commercial. Cross-pool generalization is shown, since the result replicates on top-five men's leagues and on tournaments, women's, and other competitions (Section 4.5), but cross-dataset generalization on a fully independent provider remains future work. The learned-model coverage on ScoutBench includes a faithful published Player-Vectors baseline that ties raw, v11, a purpose-trained metric learner, NMF, and card+VAEP, all of which fail to beat raw; RisingBALLER as a transformer foundation model remains future work. On the 360 task, training with cross-match positives might learn a less-leaky encoder and partially recover signal, which is an open question rather than a refutation.

9 Conclusion

Two widely-used styles of player-representation evaluation grade the model against information that the model can see. When relevance labels come from the model's own clusters, scores saturate near 0.99 and mean nothing; when query and gallery come from the same match, an identity score of 0.878 collapses to 0.188 once the matches are made disjoint. In both cases the failure is the same: a path runs from the relevance signal back into the representation under test. The fixes differ only in where that path is cut, an external transfer anchor for label leakage and a match-disjoint split for feature leakage, and both are cheap to adopt. The discipline that ties them together is a single instruction we offer as the paper's takeaway: identify every path from the relevance signal to the representation, and sever it. Under that discipline, no learned representation we test beats a raw-statistics nearest-neighbour on all-candidates external transfer-anchored similarity — the published Player Vectors method included, and on a benchmark powered to have detected such an advantage — which relocates the open problem from better embeddings to better, externally-grounded labels.

Reproducibility

The ScoutBench code lives in `football_embed/evaluation/`, comprising `scoutbench.py`, `scoutbench_significa` `scoutbench_blockboot.py` (block bootstrap, component t-test, and Holm correction), `scoutbench_extended.py`, `scoutbench_metric_baseline.py` (multi-positive loss and TOST), `scoutbench_label_validity.py`, `scoutbench_generalization.py` (cross-pool), and `scoutbench_power.py` (minimum detectable effect / sensitivity), with the Player-Vectors builder at `football_embed/data/player_vectors_nmf.py`; results are written to `data/processed/benchmark/*.json`. The 360 protocol lives in `sb360_crossmatch_killtest` (split-half, cross-match, same-match control, and no-teammate conditions) and `sb360_6mapnet_crossmatch.py` (the static heatmap family), with data in `sb360_sets_matchkeyed.npz` (offline-rebuilt and validated) and results in `sb360_crossmatch_killtest.json`. The card matrix is sanitized (clipped to ± 50 and passed through nan-to-num), and aggregate metrics are confirmed unchanged; similarity matrices are verified finite and bounded ($|\cos| \leq 1.0$ by direct checks). A spurious BLAS matmul runtime warning appears on some macOS Arrow and Accelerate builds but does not affect results, and is suppressed at the call site where this was verified.

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